# CSE 429: Computer Vision and Pattern Recognition

Egypt-Japan University of Science and Technology (E-JUST)
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## Project Proposal: AI-Generated Image Detection

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## Problem Statement

The rapid proliferation of AI-generated images from models such as GANs and Latent Diffusion Models has created an urgent need for reliable detection systems. The core challenge is **cross-generator generalisation**: a detector trained on images from one generator (e.g., ProGAN) typically fails when tested on images from a different generator family (e.g., Stable Diffusion).

**Goal:** Given any input image, the system outputs a binary prediction — *Real* (naturally captured photograph) or *Fake* (AI-generated) — along with a confidence score. The system should generalise across generator families not seen during training.

## Approach

We re-implement **UniversalFakeDetect** (Ojha et al., CVPR 2023), which addresses cross-generator generalisation by performing classification in the feature space of a frozen CLIP-ViT-L/14 encoder.

**Architecture:**

- **Backbone:** Frozen CLIP-ViT-L/14 visual encoder (307M parameters, no gradient updates)
- **Head:** Single trainable Linear(768→1) layer — only 769 parameters trained
- **Loss:** Binary cross-entropy with logits
- **Optimiser:** AdamW (lr=0.001, weight decay=0.001), Cosine Annealing LR, early stopping

The key insight is that CLIP features capture universal visual semantics without any prior exposure to the real-vs-fake distinction, avoiding the "sink class phenomenon" that causes prior methods to fail on unseen generators.

## Experiments and Results Plan

**Datasets:**

- **CIFAKE** — 60K real (CIFAR-10) + 60K Stable Diffusion v1.4 images
- **Real and Fake Face Detection** — Real photographs + StyleGAN-generated faces
- Train/Val/Test split: 70%/15%/15% (stratified, seed=42)

**Baselines:**

- Random Classifier (chance-level baseline)
- Simple CNN trained from scratch (4-layer convolutional network)

**Evaluation Metrics:** AUC-ROC, Accuracy, F1-Score, Average Precision (AP)

**Success Criteria:** AUC $> 0.90$ on the combined test set, demonstrating that CLIP-based linear probing outperforms both baselines and generalises across generator types.

**Implementation:** PyTorch + open-clip-torch. CLIP backbone loaded from pretrained OpenAI weights. Training on NVIDIA RTX 3050 GPU.

**Deployment:** Gradio web application deployed on Hugging Face Spaces for live demo.

## References


- U. Ojha, Y. Li, Y. J. Lee, "Towards Universal Fake Image Detectors that Generalize Across Generative Models," CVPR 2023.
- A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021.
- J. J. Bird, A. Lotfi, "CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images," arXiv:2303.14126, 2023.